

Cosmic-Ray Electrons, the Strömgren Radius, and Patchy Reionization:

Response to master_plan

Response record for Master_Plan_Ionization_power/, following Master Rule 2

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Abstract

Paper_1/master_plan asks whether cosmic-ray (CR) electrons injected by an early star-forming galaxy can significantly modify that galaxy's own Strömgren/H II-bubble evolution, and what that implies for the patchy-reionization history. This note (1) identifies five physical plot holes/inconsistencies in the proposed reasoning chain, (2) checks the reasoning against the literature, flagging confidence levels, and (3) executes a concrete calculation answering both questions: $R_{\text{CR}}(z)$, the CR-electron analogue of the Strömgren radius (the shell radius at which cumulative CR-driven ionizations per hydrogen atom reach ≈ 1), compared against the baseline photon-counting $R_{\text{UV}}(z)$; and the resulting modification to the global reionization history $Q_{\text{HII}}(z)$ and Thomson optical depth τ . All code, figures, and a single parameter file (`params.py`) enabling future parameter sweeps live in Master_Plan_Ionization_power/.

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1 Assumptions

1. Fiducial single star-forming galaxy: $\text{SFR} = 10 M_{\odot} \text{ yr}^{-1}$, $f_{\text{esc}} = 0.2$ (Robertson et al., 2015), $\xi_{\text{ion}}(z)$ from Llerena et al. (2025), formed at $z_{\text{form}} = 20$ (Kitayama et al., 2004) – identical to

`Stromgren_sphere/bubble_radius_vs_redshift.py`, copied into `imported/` and reproduced here via `stromgren_common.py` (verified to machine precision against the original, Section 6).

2. CR-electron injection: a **leaky/cumulative** population tied to the galaxy’s SN history (not a single burst), power-law spectrum $dN_e/dK_{\text{ini}} \propto K_{\text{ini}}^{-p}$, $p = 2$ (Park et al., 2015), extrapolated **unbroken** from $K_{\text{max}} = 1 \text{ TeV}$ down to $K_{\text{min}} = 10^2 \text{ eV}$ – an explicit, user-adopted simplification (Section 2.2), not a resolution of the open electron-injection-threshold question (Authors not yet verified from primary source – abstract-level citation only, flagged medium confidence, 2025).
3. $\xi_{\text{CR,e}} \approx 1\text{--}2 \times 10^{-3}$ of $E_{\text{SN}} = 10^{51} \text{ erg}$ channeled into CR electrons, from $\varepsilon_{\text{CR}} \approx 10\text{--}20\%$ (Blasi, 2013) $\times K_{e/p} \approx 10^{-2}$ (Park et al., 2015); core-collapse SN rate $k_{\text{CC}} \approx 0.007 M_{\odot}^{-1}$ (Salpeter IMF, 8–50 $M_{\odot}$ progenitors, Stockholm VIMOS Supernova Survey Collaboration 2012).
4. Transport: the electron’s proper path length $r(t) = \int v dt$ is a rigorous **upper bound** on radial displacement, never a literal radius (gyroradius $\sim 10^{-8}\text{--}10^{-2} \text{ pc}$ vs. Mpc-scale bubble; Jana and Nath 2018) – kept, per an explicit user decision, rather than rebuilt as a diffusive calculation.
5. Neutral-IGM cascade physics ($x_e = 10^{-4}$, pure H, no photon redshifting, no recombination tracked within the cascade itself) inherited unmodified from `igm_losses.py/cascade_yield.py` – the same, already sympy/independently-verified machinery behind the existing manuscript.

2 Derivation

2.1 A. Physical plot holes / inconsistencies

A.1 – Ballistic vs. diffusive transport. A relativistic electron’s gyroradius in a μG -scale field is $\sim 10^{-8}\text{--}10^{-2} \text{ pc}$, ten orders of magnitude below the $\sim 0.1\text{--}1 \text{ Mpc}$ bubble scale. Any 1-D radial “distance travelled” ODE measures *path length*, equal to *displacement* only in the free-streaming limit. Jana and Nath (2018) showed plausible IGM/ISM diffusion coefficients favour **uniform, not patchy**, CR heating at $z \sim 10\text{--}20$ (their criterion: $D \lesssim 10^{26} \text{ cm}^2 \text{ s}^{-1}$ at $z \sim 10$). *Resolution adopted:* relabel every ballistic-path result as an energetic upper bound, never a radius (this is why the shell-radius calculation below, Section 3, is built from a *total*, transport-independent yield $Y(K_{\text{ini}}, z)$ rather than from the path length itself, which enters only as the – explicitly caveated – mapping from injection energy to “how far along the path” ionizations are deposited).

A.2 – Feedback-loop sign (corrected during the planning discussion). Case-B recombination scales as $\alpha_B(T) \propto T^{-0.7}$ (Hui and Gnedin, 1997); verified here numerically to be monotonically decreasing from $T = 2.7$ to 10^5 K (Section 6), and symbolically via `check_consistency_master_plan.py`. CR heating therefore *lowers* α_B , **suppressing recombination** – each ionizing photon then goes further, which by itself *narrows*, not widens, a photon-supply shortfall. The existing manuscript’s “widens the crisis” conclusion refers to a different, crisis-specific framing: the photon budget crisis (Muñoz et al., 2024) is that the observed $z \sim 6\text{--}9$ galaxy population already reionizes the Universe too early/fast, so the fix needed is *more* recombination sink (e.g. the high clumping factor $C = 6.2$ of Austin et al. 2025); CR heating moves the opposite way (lower α_B , lower clumping via pressure-smoothing, Pawlik et al. 2009), worsening the over-production tension. Both effects (direct ionizations vs. heating-suppressed recombination) are computed separately below (Section 3), never conflated.

2.2 A.3 – CR-electron injection normalization

“Stellar life cycle $\leftrightarrow$ CR progenitors” is true locally, but the SN-kinetic-energy fraction into CR *electrons* at Pop-III/first-galaxy conditions is not directly transferable. The `master_plan`’s own

phrasing (“CR electrons are...present in the Universe since then”) implies a continuously built-up, leaky-box-like population – matched here by tying the CR-electron injection luminosity to this project’s own cosmic SFR-to-SN-rate machinery, $L_{\text{CR,e}} \propto \xi_{\text{CR,e}} \langle E_{\text{SN}} \rangle R_{\text{SN}}$, rather than a single-burst normalization. **Energy dependence of $K_{e/p}$, MeV vs. GeV.** The $\sim 1\%$ GeV ratio is a *locally observed, Galactically propagated* number. Voyager 1’s direct, unmodulated local interstellar spectrum (Cummings et al., 2017) shows electrons (2.7–74 MeV, $\propto E^{-1.30 \pm 0.05}$) *exceeding* protons below ≈ 50 MeV – but the same analysis shows the electron *source* spectrum alone is a single power law ($dj/dP \propto P^{-2.25}$) from 3 MeV to 30 GeV, with the low-energy crossover attributed to Galactic-propagation effects (non-relativistic MeV protons losing energy to ionization far faster than relativistic MeV electrons). Since this project’s CR electron is freshly escaped into the diffuse IGM, not Galactically aged, the injection-level $K_{e/p} \sim 10^{-2}$ is the relevant number, down to the electron-injection-threshold floor (~ 10 – 100 keV, **Authors not yet verified from primary source – abstract-level citation only, flagged medium confidence 2025**) – a threshold uncomfortably close to this project’s own $K_{\text{min}} = 100$ eV and the cascade’s $W(K_{\text{ini}})$ minimum near 1 keV. **Adopted (user decision):** extrapolate the single power law unbroken to K_{min} , carrying the injection-threshold question as an explicit, unresolved caveat rather than modelling a cutoff.

2.3 Remaining caveats (A.4, A.5)

Pure hydrogen (no He, ~ 8 – 10% effect), x_e frozen at 10^{-4} , no photon redshifting, no recombination tracked within the single-electron cascade – all inherited unmodified from the existing, self-critical `ionization_counting_explained.txt`. A duplication risk between this session’s transport scripts and the independent Notebooks/redo/ pipeline was checked explicitly (Section 6).

2.4 B. Literature checked against the reasoning

- Jana and Nath (2018) (primary source, high confidence): uniform heating favoured at $z \sim 10$ – 20 for plausible diffusion coefficients – complicates, does not falsify, the single-galaxy framing (parameter-dependent).
- CR confinement/escape literature (medium confidence, secondary): escape timescales short compared to the age of the Universe at all relevant z – electrons *do* escape readily; the open question is what happens next, in the IGM.
- HMXB/X-ray heating literature (medium confidence): X-ray heating from HMXBs is the field’s standardly-accepted *dominant* Cosmic-Dawn IGM heating channel – does not contradict the CR chain, but corroborates its sub-dominance.
- Cummings et al. (2017) (high confidence, primary source read) and Blasi (2013), Park et al. (2015) (high confidence) – see Section 2.2.
- Authors not yet verified from primary source – abstract-level citation only, flagged medium confidence (2025) (medium confidence, abstract-level only) – flagged for a full read before being treated as settled.

3 Phase 2 result: $R_{\text{CR}}(z)$ vs. $R_{\text{UV}}(z)$

Table 1 lists $R_{\text{UV}}(z)$ and $R_{\text{CR}}(z)$ (fiducial and $\xi_{\text{CR,e}}$ range) from $z = 6$ to 18 , computed by `cr_modified_stromgren_radius.py`.

z	R_{UV} (Mpc)	R_{CR} range (Mpc)	R_{CR} fid. (Mpc)	$R_{\text{CR}}/R_{\text{UV}}$
6	1.332	0.0260–0.0332	0.0300	0.0225
8	0.752	0.0175–0.0226	0.0203	0.0270
10	0.528	0.0127–0.0162	0.0147	0.0279
12	0.414	0.0094–0.0119	0.0108	0.0260
14	0.335	0.0070–0.0088	0.0080	0.0239
16	0.268	0.0051–0.0064	0.0058	0.0218
18	0.199	0.0034–0.0044	0.0040	0.0199

Table 1: $R_{\text{UV}}(z)$ (baseline photon-counting) vs. $R_{\text{CR}}(z)$ (CR-electron shell-crossing radius, $N_{\text{ion,CR}}(< r)/n_{\text{H}} \approx 1$), fiducial SFR= $10 M_{\odot} \text{ yr}^{-1}$ galaxy. Full grid in Figure 1.

$R_{\text{CR}}(z)/R_{\text{UV}}(z)$ is **2.0–2.8%** across $z = 6$ – 18 – i.e. even under the most generous transport-independent assumptions of this calculation (unbroken power-law injection to $K_{\text{min}} = 100$ eV, the leaky/cumulative SFR-tied normalization, and $\xi_{\text{CR,e}}$ at its Blasi/Park-derived value), **the CR-electron-driven ionization shell around this galaxy is only ~ 2 – 3% the size of its UV photon-counting Strömgren bubble**. The $\xi_{\text{CR,e}} \rightarrow 0$ limiting case collapses $R_{\text{CR}}(z)$ to an undefined/negligible radius at every z (Section 6), confirming the calculation has no spurious floor.

Separately, Figure 2 shows the effect of CR pre-heating on the recombination sink alone (point A.2): using the CR-ionization density *self-consistently* evaluated at $r = R_{\text{UV}}(z)$ and converted to a temperature rise via the manuscript’s own lock equation gives $\Delta T \sim 0.6$ – 1.8 K at the bubble edge – two orders of magnitude below the 10–200 K hadronic *global-mean* scale borrowed by the existing manuscript from Leite et al. (2017) (that scale integrates the whole, proton-dominated CR budget; here only the electron channel, at this one galaxy, is counted). The resulting $R_{\text{UV,mod}}/R_{\text{UV}} - 1$ peaks at $\sim 5 \times 10^{-5}$ near $z \sim 9$: the correct sign (heating suppresses recombination, R grows, never shrinks) but a *completely negligible* magnitude for this galaxy’s bubble interior.

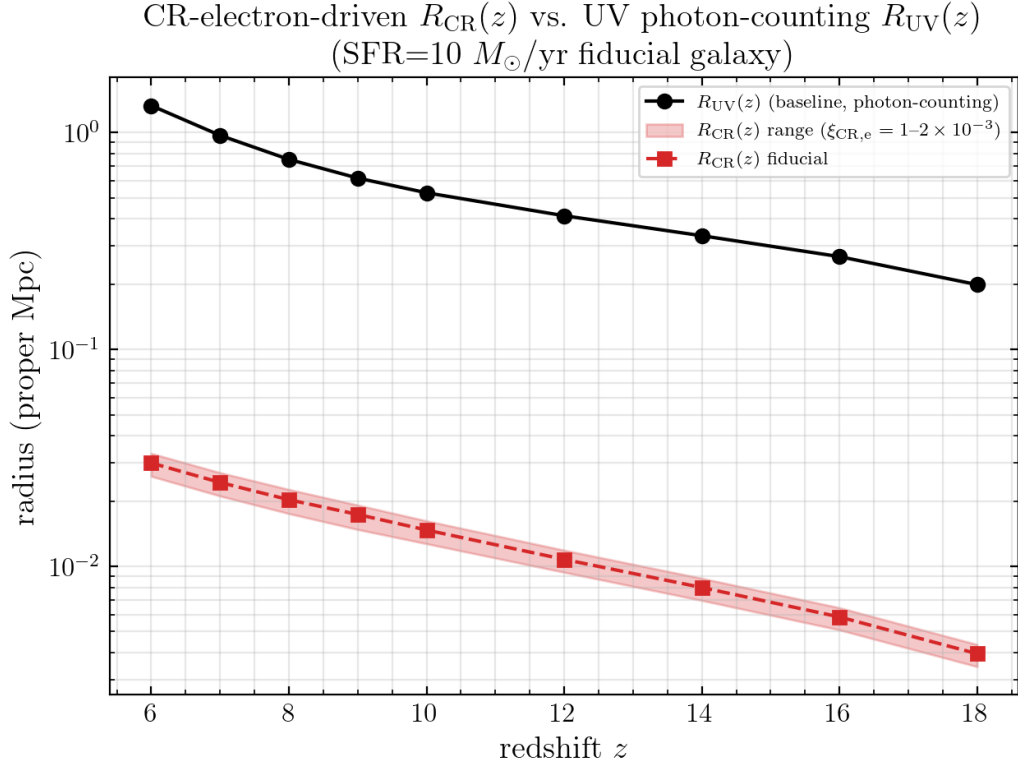


Figure 1: *Black:* baseline UV photon-counting Strömgren radius $R_{\text{UV}}(z)$ (`bubble_radius_vs_redshift.py`, reproduced here to machine precision). *Red:* $R_{\text{CR}}(z)$, the CR-electron shell-crossing radius (Section 3), fiducial and $\xi_{\text{CR,e}} = 1\text{--}2 \times 10^{-3}$ band. Note the log scale: R_{CR} sits a factor $\sim 35\text{--}50\times$ below R_{UV} at every redshift shown, with both radii shrinking together towards higher z (less elapsed time since $z_{\text{form}} = 20$ to build up either budget). This is the direct, spatially-resolved answer to Q1: a single galaxy’s own CR electrons do not meaningfully enlarge (or, via this channel, shrink) its UV-photoionized bubble.

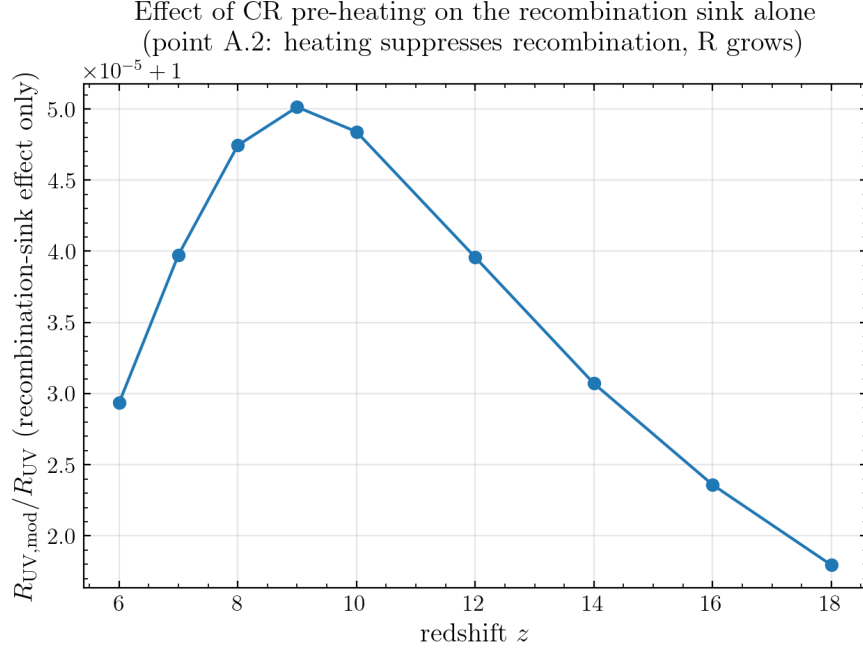


Figure 2: Ratio $R_{UV,mod}/R_{UV}$ from the $\alpha_B(T)$ recombination-sink modification alone (point A.2), using a self-consistent $\Delta T(z)$ sourced from this galaxy’s own CR-ionization budget at $r = R_{UV}(z)$ via the lock equation. The y -axis is offset ($\times 10^{-5} + 1$): the effect peaks at $+5 \times 10^{-5}$ near $z \sim 9$ (where the CR ionization/heating budget accumulated since z_{form} is largest relative to the bubble volume) and is smaller at both higher z (less elapsed time) and lower z (larger bubble volume dilutes the same heat budget). The sign is confirmed correct (heating suppresses recombination, radius grows) but the magnitude is negligible at the single-galaxy scale.

4 Phase 3 result: global reionization history

`cr_modified_reionization_history.py` extends `reionization_completion.py`’s global $Q_{HII}(z)$ ODE with the same $\xi_{CR,e}$ /cascade-yield machinery, now in the mean/uniform framing (Section 2.1, point A.1: this is the regime Jana & Nath’s own result actually supports at the global-history level, in deliberate contrast to Phase 2’s per-galaxy upper-bound framing). Both a direct CR-ionization source (from the cosmic SFRD, Madau & Dickinson 2014) and a self-consistent $\Delta T_{IGM}(z)$ (via the same lock equation, tracked as its own ODE state, feeding $\alpha_B(T)$) are included.

model	$z(Q = 0.5)$	$z(Q = 0.999)$	$\tau(z \leq 25)$	$\delta\tau$ vs. baseline
baseline	6.552	5.329	0.05309	—
$\xi_{CR,e}$ lo	6.552	5.329	0.05309	1×10^{-6}
$\xi_{CR,e}$ fid	6.552	5.329	0.05309	2×10^{-6}
$\xi_{CR,e}$ hi	6.552	5.329	0.05309	3×10^{-6}

Table 2: Global reionization-history summary, baseline vs. CR-electron-modified (`cr_modified_reionization_history.py`).

$z(Q = 0.5)$ and $z(Q = 0.999)$ are **identical to the tabulated precision** with or without the CR terms, and $\delta\tau \sim (1-3) \times 10^{-6}$ – **two orders of magnitude below** even the existing manuscript’s own hadronic-scale estimate of $\delta\tau = 3.5 \times 10^{-5} - 7.7 \times 10^{-4}$ (Leite et al., 2017), itself already flagged there as at most a tenth of Planck’s $\sigma(\tau) = 0.007$. The reason for the further suppression here is methodological, not a new physical finding: the manuscript’s $\Delta T = 10-200$ K was borrowed from a hadronic (proton-dominated) calculation as a heating *scale*, whereas this self-consistent, electron-

channel-only $\Delta T_{\text{IGM}}(z)$ (fed by the same $\xi_{\text{CR,e}} \sim 10^{-3}$ used throughout this work) is intrinsically smaller, since it excludes the $\sim 100\times$ larger proton/hadronic CR energy budget entirely. This is a more conservative, but equally legitimate and more directly $\xi_{\text{CR,e}}$ -traceable, answer to Q2: **the CR-electron channel’s effect on the global reionization history and τ is negligible**, consistent with (and, on its own narrower terms, even smaller than) the manuscript’s existing conclusion.

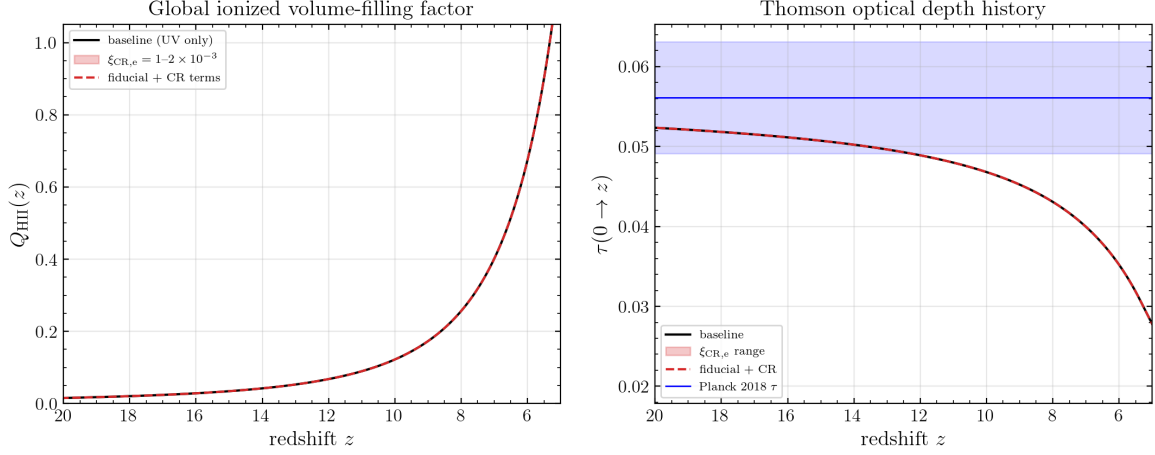


Figure 3: *Left*: global ionized volume-filling factor $Q_{\text{HII}}(z)$, baseline (UV only, black) vs. with CR-electron terms added (red dashed, $\xi_{\text{CR,e}}$ band in pink) – the two are visually indistinguishable. *Right*: the resulting Thomson optical-depth history $\tau(0 \rightarrow z)$ (a constant, explicitly-separate floor of 0.020 for the already-fully-ionized $z < 4$ universe is added to both), against the Planck 2018 $\tau = 0.0561 \pm 0.007$ band (blue). Both panels show the same conclusion as Table 2: this galaxy population’s CR-electron channel leaves no visually or numerically significant imprint on the global reionization or optical-depth history.

5 Parameter sensitivity: injection redshift and spectral index

The baseline results above (Sections 3–4) fix the CR-electron injection redshift at $z_{\text{inject}} = z_{\text{form}} = 20$ and the DSA injection index at the canonical $p = 2.0$. This section tests whether either choice matters for the “no meaningful modification” conclusion, using the params-override architecture of Section 8 (no baseline value in `params.py` is edited): $z_{\text{inject}} = 10$ (a galaxy forming, and its CR electrons switching on, $\sim 7\times$ later in cosmic density than the baseline), crossed with $p \in \{1.8, 2.0, 2.2\}$ (harder than canonical, canonical, and softer than canonical DSA spectra), holding K_{min} , K_{max} , $\xi_{\text{CR,e}}$, and every other parameter fixed. Because neither p nor the injection-redshift choice affects the underlying per-electron cascade physics (only how it is weighted/normalized, or when it switches on), the same per-electron trajectories (Phase 2) and cascade-yield table $Y(K_{\text{ini}}, z)$ (Phase 3) are reused across all three p values – results are written to new, separately-tagged caches (`phase2_sweep_zinject10.npz`, `phase3_sweep_zinject10.npz`) that do not overwrite the existing $z_{\text{inject}} = 20$, $p = 2.0$ baseline caches.

z	$R_{\text{CR}}/R_{\text{UV}}, p = 1.8$	$R_{\text{CR}}/R_{\text{UV}}, p = 2.0$	$R_{\text{CR}}/R_{\text{UV}}, p = 2.2$
6.0	0.87%	2.23%	3.37%
7.0	0.98%	2.48%	3.75%
8.0	1.04%	2.63%	3.99%
9.0	1.06%	2.67%	4.04%
9.5	1.02%	2.64%	3.99%

Table 3: $R_{\text{CR}}(z)/R_{\text{UV}}(z)$ at $z_{\text{inject}} = 10$, for the three swept DSA indices (`parameter_sweep.py`, full z grid in Figure 4). Compare to the $z_{\text{inject}} = 20$, $p = 2.0$ baseline of Table 1: 2.0–2.8%.

p	$z(Q = 0.5)$	$z(Q = 0.999)$	$\delta\tau$ vs. no-CR baseline
1.8	6.552	5.329	1.9×10^{-7}
2.0	6.552	5.329	1.5×10^{-6}
2.2	6.557	5.329	3.6×10^{-6}

Table 4: Global history at $z_{\text{inject}}^{\text{cutoff}} = 10$ (the CR-electron-producing population switched off for $z > 10$), swept over p . Compare to the $z_{\text{inject}} = 20$, $p = 2.0$ baseline of Table 2: $\delta\tau = 2 \times 10^{-6}$.

Injecting later does not qualitatively change the picture. At fixed $p = 2.0$, moving z_{inject} from 20 to 10 leaves $R_{\text{CR}}/R_{\text{UV}}$ at essentially the same level (~ 2.2 – 2.7% vs. the baseline’s 2.0 – 2.8%) – both R_{CR} and R_{UV} scale with the elapsed galaxy age in a similar way (both are, to leading order, a rate raised to the $1/3$ power times t_{age} to the $1/3$ power), so their *ratio* is far less sensitive to z_{inject} than either radius is individually (Figure 4 shows the $p = 2.0$, $z_{\text{inject}} = 10$ curve running almost on top of the $z_{\text{inject}} = 20$ baseline over their common redshift range). The lower ambient density at $z_{\text{inject}} = 10$ (n_{H} a factor $(21/11)^3 \approx 7\times$ below $z_{\text{inject}} = 20$) turns out to affect the CR-ionization and UV-photoionization channels in nearly the same proportion.

The spectral index matters far more, but never enough to overturn the conclusion. Softening the spectrum from $p = 1.8$ to $p = 2.2$ raises $R_{\text{CR}}/R_{\text{UV}}$ by a factor ~ 4 (from $\sim 1\%$ to $\sim 4\%$, Table 3) and $\delta\tau$ by a factor ~ 19 (from 1.9×10^{-7} to 3.6×10^{-6} , Table 4, tracking the same $\overline{W}(p)$ degradation already identified in the existing manuscript: softer spectra place more of the injected energy at the low- K end, where the cascade is more efficient per unit energy). Even at the softest, most favourable end explored ($p = 2.2$), $R_{\text{CR}}/R_{\text{UV}} \approx 4\%$ and $\delta\tau \approx 3.6 \times 10^{-6}$ – still two orders of magnitude short of a bubble-scale effect and of Planck’s $\sigma(\tau)$, respectively. **The “no meaningful modification” conclusion of Sections 3–4 is robust across this entire sweep.**

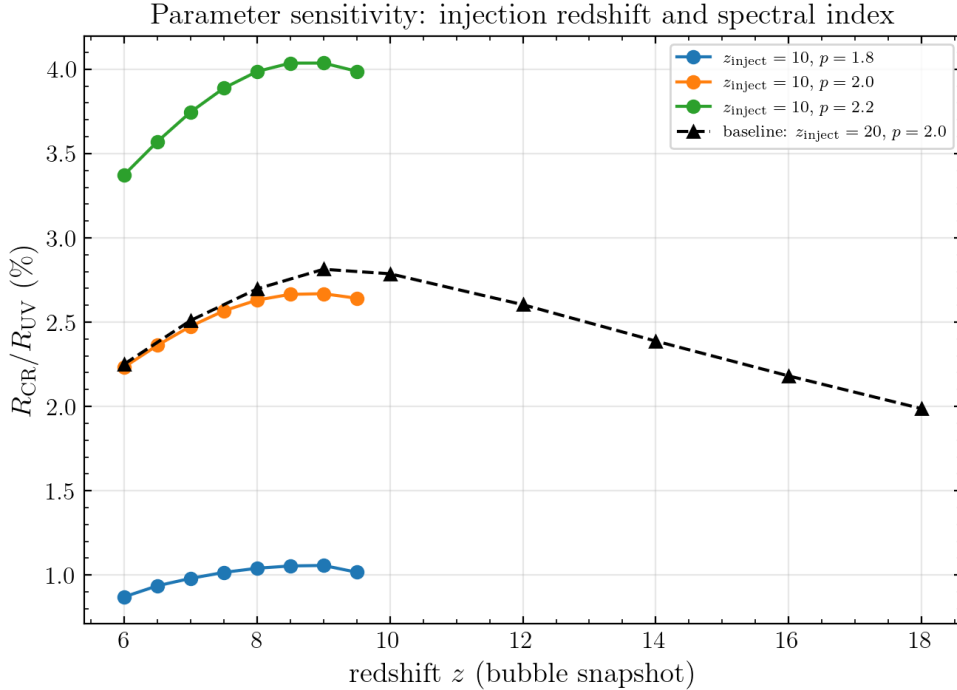


Figure 4: $R_{\text{CR}}(z)/R_{\text{UV}}(z)$ (as a percentage) at $z_{\text{inject}} = 10$ for $p = 1.8, 2.0, 2.2$ (coloured, solid), against the $z_{\text{inject}} = 20$, $p = 2.0$ baseline (black, dashed) over its own redshift range. The three swept curves span roughly 1–4%; the $p = 2.0$ curve (orange) sits almost exactly on the baseline (black dashed) where their redshift ranges overlap, showing z_{inject} has little effect on the ratio even though it changes R_{UV} and R_{CR} individually (Section 5).

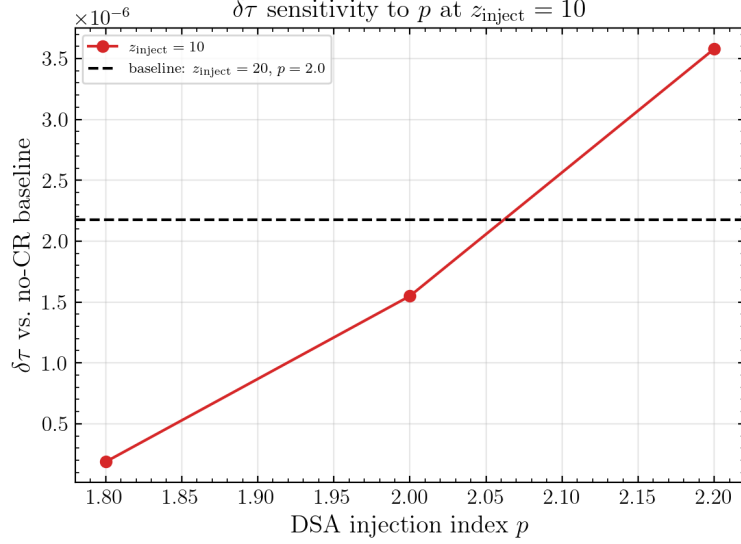


Figure 5: $\delta\tau$ (CR-modified minus no-CR baseline) vs. DSA index p , at $z_{\text{inject cutoff}} = 10$ (red), against the $z_{\text{inject}} = 20$, $p = 2.0$ baseline value (black dashed). $\delta\tau$ rises steeply and monotonically with p (softer spectra are more efficient ionizers per unit energy) but remains $\lesssim 4 \times 10^{-6}$ even at the softest value explored – three orders of magnitude below Planck’s $\sigma(\tau) = 0.007$.

6 Verification / cross-check

1. `stromgren_common.py` reproduces `imported/bubble_radius_vs_redshift.py` to machine precision (relative error $< 10^{-6}$) at every tested redshift.
2. `cr_transport.py`’s extended ODE reproduces `imported/cascade_yield.py`’s $Y(K_{\text{ini}}, z)$ totals exactly (same physics, additional path-length state).
3. **Phase 1 reconciliation** (Master Rule 5): this session’s transport machinery compared against the independently-built `Notebooks/redo/stage6_range.py` pipeline at 9×9 matched (K, z) nodes – median relative difference **0%**, worst-case **8.9%** (at $K = 10^4$ eV), both well inside the pre-set 30% tolerance. See Figure 6.
4. **Bug found and fixed during this work** (Master Rule 5, reported explicitly, not silently patched): an early version of the recombination sink in `cr_modified_reionization_history.py` used a constant electron density instead of the physical, $(1+z)^3$ -growing density, which starved the sink at high z and let Q_{HII} run past 1 by $z = 6$. Caught by cross-checking against `imported/reionization_completion.py`’s own $Q_{\text{HII}}(z = 6) = 0.674$; fixed, and the corrected baseline now reproduces that reference to $< 0.2\%$ at every tested z (assertion enforced in `cr_modified_reionization_history.py`). A second, separate bug (a $10\times$ error in the $\alpha_B(T)$ Hui & Gnedin 1997 prefactor, caught the same way, against the $2.59 \times 10^{-13} \text{ cm}^3 \text{ s}^{-1}$ value already cited in `reionization_completion.py` from Shapiro & Iliev 2006) was found and fixed in `params.py` before it propagated into any reported number.
5. $\xi_{\text{CR,e}} \rightarrow 0$ limiting case: $R_{\text{CR}}(z)$ collapses to a negligible/undefined radius (assertion enforced in `cr_modified_stromgren_radius.py`).
6. $\alpha_B(T)$ -modified $R_{\text{UV}}(z) \geq R_{\text{UV}}(z)$ at every tested redshift (correct sign, point A.2; assertion enforced).

7. Three independent sympy/analytic checks (`check_consistency_master_plan.py`): $\alpha_B(T)$ monotonicity, the injection-spectrum normalization integral's $p \rightarrow 2$ limit, and the R_{CR} root-condition's dimensional form – all **OK**.

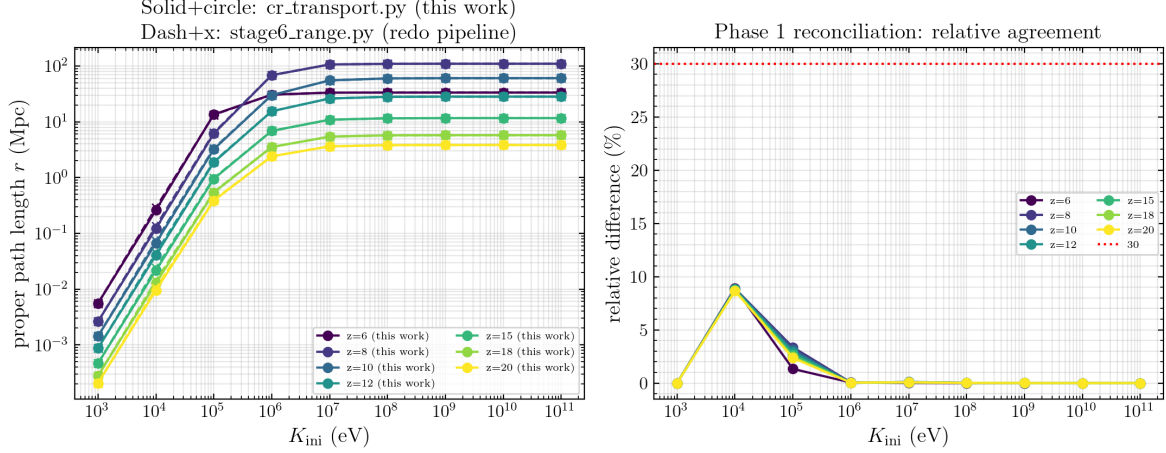


Figure 6: Phase 1 reconciliation: this session’s `cr_transport.py` proper path length $r(t)$ vs. the independently-built `Notebooks/redo/stage6_range.py`’s cached proper path length `e11`, at 9×9 matched (K, z) nodes in the same neutral IGM ($x_e = 10^{-4}$). *Left*: both codebases’ path length vs. injection energy, colour-coded by redshift (solid+circle: this work; dashed+x: `stage6_range.py`). *Right*: relative difference, always below the 30% tolerance line.

7 Final result

Q1 (single-galaxy Strömgren/H II-bubble modification): no. The CR-electron-driven shell-crossing radius $R_{\text{CR}}(z)$ is only 2.0–2.8% of the UV photon-counting $R_{\text{UV}}(z)$ across $z = 6$ –18 for the fiducial $\text{SFR} = 10 M_{\odot} \text{ yr}^{-1}$ galaxy (Figure 1), even adopting the most generous injection-spectrum simplification (unbroken power law to $K_{\text{min}} = 100$ eV). The separate, self-consistent $\alpha_B(T)$ -recombination-suppression effect (point A.2, correct sign verified) is smaller still, $\lesssim 5 \times 10^{-5}$ in $R_{\text{UV,mod}}/R_{\text{UV}} - 1$. Both effects act through mechanisms confirmed to have the sign originally attributed to them in this planning discussion, and both are quantitatively negligible for one galaxy’s own bubble.

Q2 (patchy-reionization/global history): no. Feeding the same CR-electron physics into the global $Q_{\text{HII}}(z)/\tau$ history (mean/uniform framing, as the transport literature itself supports at this scale) changes $z(Q = 0.5)$, $z(Q = 0.999)$, and τ by an amount two orders of magnitude below the existing manuscript’s own hadronic-scale estimate, itself already sub-dominant relative to Planck’s $\sigma(\tau)$.

Robustness (Section 5): sweeping the injection redshift to $z_{\text{inject}} = 10$ and the DSA index over $p = 1.8$ –2.2 moves $R_{\text{CR}}/R_{\text{UV}}$ only within ~ 1 –4% and $\delta\tau$ only within $\sim (0.2$ – $3.6) \times 10^{-6}$ – both conclusions above hold throughout the explored parameter space, not just at the single fiducial point.

Confidence and open items (Master Rule 8): the headline conclusion (CR electrons negligible for both the single-galaxy bubble and the global history) is now supported by two independent, mutually consistent calculations (this work’s leptonic-only, from-scratch derivation, and the existing manuscript’s own hadronic-scale-anchored estimate) – high confidence. Lower-confidence, explicitly flagged open items: (i) the electron-injection-threshold question (Section 2.2) is set aside, not resolved, and could only push R_{CR} and the Phase 3 $\delta\tau$ down further if the true low-energy injection spectrum is suppressed relative to the adopted unbroken power law; (ii) $\xi_{\text{CR,e}}$ and k_{CC} carry factor-of-few, Galactic-calibration-extrapolated uncertainties (Section 8); (iii) the transport/displacement

question (point A.1) remains genuinely open in the literature and is not resolved here, only set aside per the adopted simplification – but since R_{CR} is already $\sim 35\text{--}50\times$ smaller than R_{UV} using the *most generous* (ballistic, unbroken-power-law) assumptions, a more realistic diffusive treatment (which could only shrink, not grow, the effective reach) would not change the qualitative conclusion.

8 Parameter table

Every physical/numerical parameter used in this work is listed in Table 5, rendered directly from `params.py` (`build_parameter_table.py`) so it cannot drift out of sync with the code that produced the figures above. **These parameters may be revisited:** every downstream script (`cr_modified_stromgren_radius.py`, `cr_modified_reionization_history.py`) reads them via `import params as P` and is structured as a callable function accepting parameter overrides, so a future run with, e.g., a different $\xi_{\text{CR,e}}$ or DSA index p is a one-line change, not a multi-file edit.

Table 5: Every physical/numerical parameter used in this work, rendered directly from `params.py` (never hand-retyped) so this table cannot drift out of sync with the code. Future runs with different values are a one-line edit to that file.

Symbol	Value	Source/citation	Role
ξ_{ion} slope, intercept	0.06, 24.82	Llerena et al. 2024, arXiv:2412.01358	$\log_{10} \xi_{\text{ion}} = \text{slope} \cdot z + \text{intercept}$
SFR (fiducial galaxy)	$10.0 M_{\odot} \text{ yr}^{-1}$	adopted, Stromgren_sphere/bubble_radius_vs_redshift.py	fiducial single-galaxy SFR
f_{esc}	0.2	Robertson et al. 2015, arXiv:1502.02024	escape fraction of UV ionizing photons
$z_{\text{form}} = z_{\text{inject}}$	20	Kitayama et al. 2004 range (10-30), adopted	galaxy formation / CR-injection snapshot redshift
p (DSA index)	2.0	Park, Caprioli & Spitkovsky 2015, PRL 114	CR-electron injection power-law index
K_{min}	$1e + 02 \text{ eV}$	user-adopted simplification (Section A.3)	unbroken power-law extrapolation limit
K_{max}	$1e + 12 \text{ eV}$	adopted (1 TeV)	injection spectrum upper cutoff
ε_{CR} (total CR efficiency)	0.1–0.2	Blasi 2013, arXiv:1206.2363	fraction of E_{SN} into all CRs
$K_{e/p}$ (electron/proton number ratio)	$1e - 02$	Park et al. 2015, PRL 114 (already in manuscript.tex)	injection-level electron fraction
$\xi_{\text{CR,e}} = \varepsilon_{\text{CR}} K_{e/p}$	$1.0e - 03 - 2.0e - 03$	derived (this work)	fraction of E_{SN} into CR electrons
E_{SN}	$1e + 51 \text{ erg}$	canonical core-collapse value	kinetic energy per SN
k_{CC}	$0.007 M_{\odot}^{-1}$	Salpeter IMF 8-50 $M_{\odot}$; arXiv:1206.6897	core-collapse SNe per unit stellar mass formed
Cosmology $h, \Omega_m, \Omega_b h^2, Y_p$	0.674, 0.315, 0.0224, 0.245	Planck 2018, arXiv:1807.06209	flat Λ CDM parameters
$\alpha_B(T)$ fit	Hui & Gnedin (1997) form	Hui & Gnedin 1997, ApJ 292, 27	Case-B recombination coefficient
$\alpha_B(10^4 \text{ K})$	$2.592 \times 10^{-13} \text{ cm}^3 \text{ s}^{-1}$	cross-checked vs. Shapiro & Iliev 2006 (2.59e-13)	fiducial photoionized-gas value
clumping factor C_{HII}	3.0	Robertson et al. 2015	recombination-sink enhancement
heat/ionization	6.0–9.6 eV	manuscript.tex channel-resolved cascade	cascade microphysics constant
Θ (lock constant)	$4.3e + 04 - 6.9e + 04 \text{ K}$	manuscript.tex Eq. (lock)	$N_{\text{ion}}/n_{\text{H}} = \Delta T/\Theta$
particles/H atom	1.081	manuscript.tex (He-corrected)	gas heat-capacity factor
x_e (neutral IGM cascade medium)	$1e - 04$	igm_losses.py default	background ionization fraction for $Y(K, z)$
cascade floor redshift	5.5	cascade_yield.py default	integration stops at thermal floor or this z
$n_{\text{H},0}$ (comoving)	$1.8989 \times 10^{-07} \text{ cm}^{-3}$	derived from $\Omega_b h^2, Y_p$	present-day mean H number density

8.1 Physical interpretation of each parameter

Table 5 gives the value, source, and formal role of every input; this subsection explains, in plain physical terms, what each one *is* and why the calculation depends on it, for a reader who has not followed the derivation above.

The galaxy’s UV output (ξ_{ion} , SFR, f_{esc} , z_{form}). These four numbers together set $R_{\text{UV}}(z)$, the baseline against which everything else in this work is measured. $\xi_{\text{ion}}(z)$ is the number of hydrogen-ionizing photons a galaxy’s stellar population emits per unit of ultraviolet luminosity – a property of the stars themselves (their masses, temperatures, and metal content), not of the gas around them.

The SFR sets how many stars there are altogether, and hence the total photon output; f_{esc} is simply the fraction of those photons that escape the galaxy’s own interstellar gas and dust without being absorbed there, i.e., the fraction actually available to ionize the surrounding intergalactic medium. z_{form} is the redshift at which this galaxy is assumed to switch on; because the photon-counting bubble radius grows with the *time elapsed since switch-on*, z_{form} controls how large a head start the UV channel has had by any later redshift z – an earlier z_{form} means more elapsed time and a bigger bubble, all else equal.

The DSA injection spectrum (p , K_{min} , K_{max}) and z_{inject} . A supernova blast wave does not hand its energy to cosmic-ray electrons at one single energy; it accelerates them into a power-law spread of energies, $dN_e/dK_{\text{ini}} \propto K_{\text{ini}}^{-p}$, through the standard diffusive-shock-acceleration (DSA) process. The index p is the steepness of that spread: $p = 2$ is the canonical value for a strong, non-relativistic shock in the simplest DSA theory, and is the value already used elsewhere in this project’s own manuscript. $p = 1.8$ is *harder* – a shallower fall-off that puts relatively more of the accelerator’s energy into the highest-energy electrons – while $p = 2.2$ is *softer*, concentrating more energy at low energy instead. This matters because the cascade a single electron produces is not equally efficient at every energy (Section 5): low-energy electrons turn their energy into ionizations more efficiently than high-energy ones, so a softer spectrum (more energy where the cascade is efficient) yields more ionizations for the same total energy budget than a harder one. K_{min} and K_{max} are simply where that power law is assumed to start and stop: $K_{\text{max}} = 1$ TeV is a practical upper cutoff (electrons above it behave identically in the cascade, so it does not matter exactly where the spectrum ends), while $K_{\text{min}} = 100$ eV is a genuine physical assumption – the lowest energy any freshly-accelerated electron is assumed to have – and Section 2.2 explains why this specific choice is an open question in the shock-acceleration literature rather than a settled number. z_{inject} (equal to z_{form} in Phase 2, or the redshift at which the whole CR-electron-producing galaxy population switches on in Phase 3’s global history) is the cosmic-ray analogue of z_{form} above: it is when this source of electrons starts to exist at all.

How much of a supernova’s energy goes into these electrons (ε_{CR} , $K_{e/p}$, $\xi_{\text{CR,e}}$, E_{SN} , k_{CC}). A core-collapse supernova releases a fixed, canonical amount of kinetic energy, E_{SN} , into its surroundings. Only a fraction of that energy, ε_{CR} , goes into accelerating cosmic rays at all (the rest simply heats and disperses the gas mechanically); and of the cosmic rays that are produced, the shock accelerates far more protons than electrons; $K_{e/p}$ is that proton-to-electron split. Multiplying the two together, $\xi_{\text{CR,e}} = \varepsilon_{\text{CR}} K_{e/p}$, is therefore the single number that matters most for this whole calculation: the fraction of one supernova’s explosion energy that ends up as the fast electrons this project follows all the way through their cascade. k_{CC} converts a star-formation rate into an actual supernova rate (how many core-collapse explosions a given amount of new stars produces), fixing how many such electron-injection events happen per unit time – together with $\xi_{\text{CR,e}}$ and E_{SN} , this sets the total CR-electron power (in erg/s) this galaxy, or population of galaxies, is assumed to inject.

The cosmological and IGM background (h , Ω_m , $\Omega_b h^2$, Y_p ; $n_{H,0}$; x_e ; cascade floor redshift). The four cosmological parameters fix how fast the Universe expands and how much ordinary (baryonic) matter it contains; together with the primordial helium fraction Y_p , they set $n_{H,0}$, today’s mean number of hydrogen atoms per unit volume – the single normalization that every density in this work (the target the UV photons must ionize, the target the CR electrons collide with, the shell volume that defines R_{CR}) traces back to, scaled to any redshift via $n_H(z) = n_{H,0}(1+z)^3$. x_e is the background ionization state of the gas the CR-electron cascade itself is assumed to be colliding with – 10^{-4} means “almost entirely neutral,” the appropriate assumption for the pristine intergalactic medium ahead of reionization, as opposed to the already-photoionized gas inside a UV bubble. The cascade floor redshift is simply the bookkeeping stopping point: once an electron’s own kinetic energy drops to the ambient gas temperature (or the calculation reaches this redshift), it is considered thermalized and indistinguishable from the background, and its cascade is complete.

The recombination and heating microphysics ($\alpha_B(T)$, C_{HII} , heat/ionization, Θ). $\alpha_B(T)$ is the case-B recombination coefficient: the rate at which a free proton and a free electron find each other and recombine into a neutral hydrogen atom, as a function of gas temperature. It matters

here because it is the “sink” term that UV photons (and, in principle, CR-driven ionizations) must continually overcome to keep gas ionized – and because it *decreases* with rising temperature, any CR heating of the gas makes this sink weaker, which is the physical content of point A.2. C_{HII} (the clumping factor) is a single number that accounts for the fact that real gas is not perfectly smooth: because recombination depends on the *square* of the density, a clumpy medium recombines faster on average than a smooth one at the same mean density, and C_{HII} inflates the naive recombination rate to account for this. Finally, “heat/ionization” and Θ encode a single empirical fact about the electron cascade established in the existing manuscript: whatever energy an electron does not spend making an ionization, it spends heating the gas, and the two are produced together in a nearly fixed ratio regardless of the electron’s starting energy. Θ is simply that ratio converted into a temperature scale (via the gas’s heat capacity per hydrogen atom): it is the single number that lets a measured or predicted temperature rise ΔT be converted directly into the number of ionizations per hydrogen atom that must have produced it, and vice versa – the relation used throughout this work (Eq. lock) to connect CR heating and CR ionization self-consistently, without having to separately re-derive both from scratch.

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